

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination April 2016

PH211 Pharmaceutical Analysis-II

Date: 09/05/2016, Monday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following

- A What is potentiometric titration? Discuss the methods to determine end point in potentiometric titration. 10
- B [1] Explain Kohlraush Law. 04
[2] Explain the factors affecting conductance of a solution. 06
- C [1] Explain the various types of currents generated in Polarography with the help of a polarogram. 05
[2] Explain the working of Dropping Mercury Electrode along with a neat labeled diagram 05

Q 2 Attempt any **TWO** of the following

- A [1] Classify electrodes used in potentiometry. 03
[2] Discuss the Glass Electrode with Suitable diagram. 07
- B What is biamperometry? Discuss the basic principle and method of biamperometric titration. 10
- C [1] Describe advantages and limitations of instrumental analytical methods. 04
[2] Draw schematic diagram of saturated calomel electrode. 06

SECTION- II

Q 3 Attempt any **TWO** of the following

- A Enlist the steps involved in Thin layer Chromatography. 10
- B [1] Give the detailed classification of Chromatographic Techniques 05
[2] Explain the Plate Theory of Chromatographic Techniques. 05

C Explain the Van Deemter Equation of Chromatographic Technique. 10

Q 4 Attempt any TWO of the following

A Explain in short 10

(a) Retention Time

(b) HETP

(c) Tailing Factor

(d) Plane Polarized Light

(e) Chamber saturation

B [1] State the principle of Differential Thermal Analysis. 02

[2] Draw and explain the various peaks generated in a Differential Thermal Analysis Curve (DTA) curve. 04

[3] Draw and explain the instrumentation of Differential Thermal Analysis. 04

C [1] Write short note on Thermo Gravimetric Analysis. 06

[2] What are the significant characteristics of a thermobalance used in TGA? 04
